CSIT314 Software Development Methodologies

Review



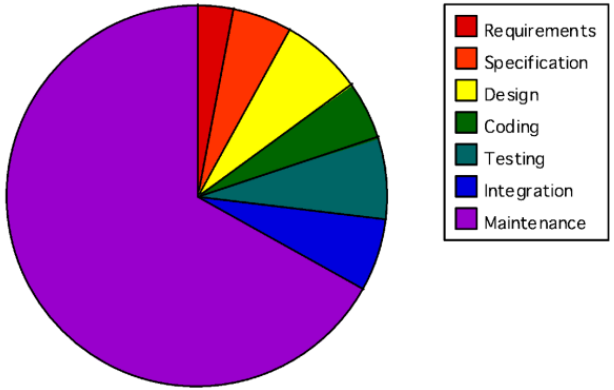
UNIVERSITY
OF WOLLONGONG
AUSTRALIA

Subject review

UNIVERSITY
OF WOLLONGONG
AUSTRALIA

Software Development

- Introduction to Software Development and its Lifecycle
  - Software Engineering
  - Software Development Activities
  - Change propagation
  - Eight laws of software evolution

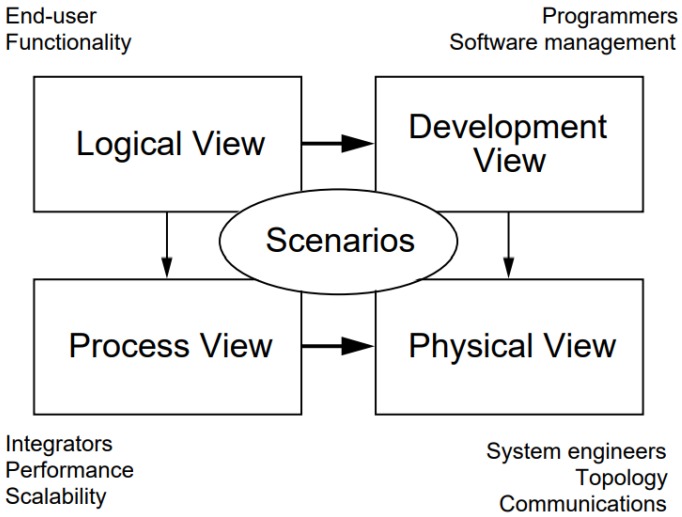


Models

- Software process models
  - Waterfall model
  - Prototyping model
  - Iterative/incremental model
  - Spiral model
  - Rational Unified Process
  - Agile methods

UML

- Unified Modelling Language



- "4+1" View
  - 1 is for user case view, or known as scenario, (user requirement)
- Conceptual and Physical
- Each view for several diagrams

2V

- Verification & Validation
  - ″build the product right“ VS “build the right product”
  - Static and Dynamic Verification
    - Software inspections / (unit, white, black-box) testing
- Test-driven development

CI & CD

- Continuous Integration
  - 9 principles and practices
- Continuous Delivery
- CI/CD with practical tools

DevOps

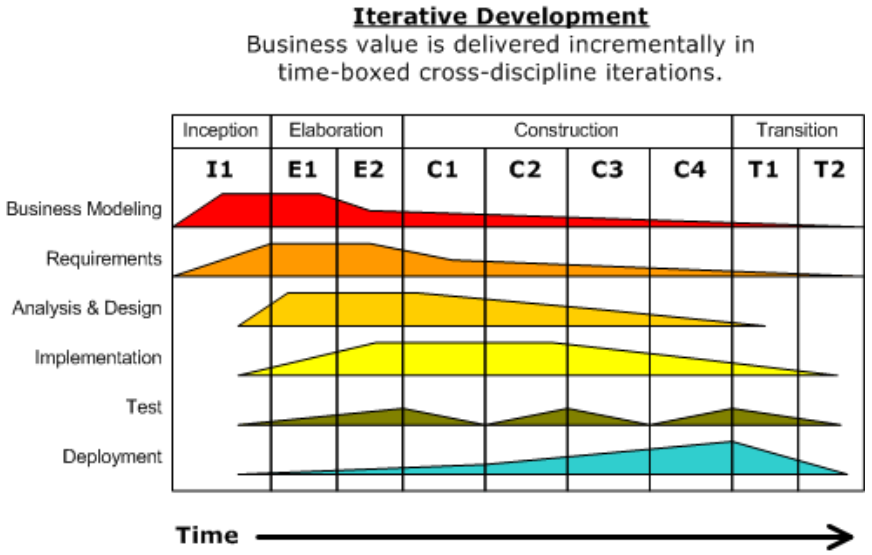
- Unify software development (Dev) and IT operation (Ops)
- DevOps Vs CI/CD, Agile

Kanban & XP

- Kanban
  - Represents the overall project (visual interface)
  - 5 components
- Extreme programming
  - 12 practices
  - Extreme Programming Process

RUP

- Rational Unified Process
  - Two ways:
    - horizontal/vertical
  - Building blocks



CMMI

- Capability Maturity Model Integration
  - 5 Maturity Levels
  - Process Areas

Maturity Level	Project Managment	Engineering	Process Management	Support
5 Optimizing			Organizational Innovation & Deployment	Causal Analysis & Resolution
4 Quantitatively Managed	Quantitative Project Mngt		Organizational Process Performance	
3 Defined	Integrated Project Mngt Risk Management	Requirements Development Technical Solution Product Integration Verification Validation	Organizational Process Focus Organizational Process Definition Organizational Training	Decision Analysis & Resolution
2 Managed	Project Planning Project Monitoring & Control Supplier Agreement Mngt	Requirements Mngt		Measurement & Analysis Process & Product Quality Assurance Configuration Mngt
1 Initial				

Data-driven

- Data-driven Software Development
  - Model requirements/deployment/monitoring
  - Data collection, cleaning, labelling and feature engineering
  - Model training, evaluation